

Web App Development 1

Computing and Mathematics

May, 2026

Web App Development 1 (A13641)

Short Title:	Web App Development 1	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Software and Web Development	
Author	EDELEASTAR	
Credits	5	Level: Introductory

Description of Module / Aims

Introduce the the student to the software development lifecycle via the implementation of a simple but functional web application. In doing this, analyse & model a constrained set of user requirements. Then design, build and deploy a simple web application. Incorporate basic database, session support & server side rendering.

Programmes

COMP-0597	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	3 / 6 / M
COMP-0597	BSc (Hons) in Applied Computing (International) (WD_KACMI_B)	3 / 5 / M
COMP-0597	BSc (Hons) in Applied Computing (WD_KACCM_B)	1 / 2 / M
COMP-0597	BSc (Hons) in Applied Computing (WD_KCOMP_B)	1 / 2 / M
COMP-0597	BSc (Hons) in Computer Forensics and Security (WD_KCOFO_B)	1 / 2 / M
COMP-0597	BSc (Hons) in Computer Science (WD_KCMSC_B)	1 / 2 / M
COMP-0597	BSc (Hons) in Creative Computing (WD_KCRCO_B)	2 / 4 / M
COMP-0597	BSc (Hons) in Software Engineering (WD_KDEVP_BI)	2 / 4 / M
COMP-0597	BSc (Hons) in Software Systems Development (WD_KDEVP_B)	2 / 4 / M
COMP-0597	BSc (Hons) in the Internet of Things (International) (WD_KINTT_BI)	3 / 6 / M
COMP-0597	BSc in Information Technology (WD_KINFT_D)	2 / 4 / M
COMP-0597	BSc in Multimedia Applications Development (WD_KMULA_D)	2 / 4 / M
COMP-0597	BSc in Software Systems Development (WD_KCOMC_D)	2 / 4 / M

Indicative Content

- User Stories & Agile context
- Introduction to Modelling
- Hypertext Transfer Protocol (HTTP) Request/Response Life Cycle
- Introductory Web Application Frameworks
- Simple Object Relational Mapping tools
- Test Driven Development

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the key components of a server rendered web application and incorporate them into a running application.
2. Use Model View Controller & related patterns in the implementation of a web project.
3. Relate the request/response lifecycle, routing & session management in the context of a modern application framework.
4. Convert a set of requirements into a set of discrete stories and translate these stories into a simple project plan with associated timeline and testing strategy.
5. Model the user requirements and realize the model in a simple database.

Learning and Teaching Methods

- Lectures will introduce the general context of the curriculum, and explore specific topics in depth.
- Supervised, guided and scripted practicals will lead the student through the construction of selection of small applications, designed to illustrate key concepts covered in the lectures.
- Each practical will propose a set of exercises -- to be solved in a subsequent practical.
- Assignment One will focus on the enhancement of the guided application.
- Assessment Two will invite the student to analyse, design and implement a basic but functional new application.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	50%	1,2,3
Assignment	50%	3,4,5

Assessment Criteria

<40%: Unable to interpret, describe or implement key components of a simple web application.

40%-49%: Be able to construct a minimal web application, incorporating server side rendering.

50%-59%: Design and implement a web application including basic database and session support.

60%-69%: All of the above including a simple model of the application structure.

70%-100%: All the above to an excellent level. Incorporate Unit Tests for a specific subset of an application.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Henderson, M. *_Instant CakePHP Starter_*. New York: Packt Publishing, 2013.
- Richard-Foy, J. *_Play Framework Essentials_*. New York: Packt Publishing, 2014.

- Syed, B. _Beginning Node.js_. New York: Apress, 2014.

Requested Resources

- COMPUTER LAB: BYOD Lab